

COMMON COURSE ASSIGNMENT FORMAT

Alternate Team Perspective - Regional Multi-Category Marketplace Edition

A. Assignment Information

Department	BTECH - AI - ML
Programme	Bachelor of Technology
Course Code & Course Name	ITA0401 - STATISTICS WITH R PROGRAMMING
Academic Year / Batch	2026-27
Faculty Name	Dr.J.Vignesh Janarthanan
Assignment Title	E-Commerce Order and Customer Analytics System
Date of Issue	02-09-2026
Date of Submission	02-09-2026
Maximum Marks	100
Course Outcome(s) - CO	CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively. Identify and implement appropriate control structures to solve a particular programming problem. CO2: Implement looping constructs, control structures, and recursive functions in R to solve computational problems with efficient program logic. Analyse the correlation between the parameters in a dataset and establish the analysis. (2,3,4) CO3: Construct and manipulate data frames using reshaping, merging, and file I/O operations in R to prepare structured datasets for analysis. CO5: Develop data visualisations and regression models in R using plotting libraries and machine learning techniques including decision trees to interpret and present analytical insights.
Bloom's Taxonomy Level	L3 - Apply; L4 - Analyse
SDG Mapping (SDG 1-17)	SDG 8 - Supports decent work and inclusive economic growth by giving small regional sellers a structured digital order and revenue pipeline.

	SDG 9 - Encourages resilient, technology-driven infrastructure for multi-category order and inventory management across Tier-2/3 markets. SDG 12 - Supports responsible consumption and production through low-stock monitoring, discount governance and reduced overstocking.
Industry / Societal Relevance	Regional & Tier-2/3 E-Commerce Multi-Category Retail Analytics Seller & Inventory Intelligence Quarterly Business Performance Reporting
Team Members	1. P.Mahendra Reddy (192425261) 2. G.Uday Siva Sai (192425259) 3.S.Sharon Surya (192324246)

B. Assignment Problem / Challenge

A regional multi-category marketplace that connects local sellers across Tier-2 and Tier-3 cities needs a dependable R-based system for tracking customers, products, inventory and orders while turning quarterly transaction records into seller- and category-level business decisions.

The challenge is to combine R programming fundamentals, object-oriented representation, reusable functions, data-frame operations, file handling and statistical visualisation into a single menu-driven analytics workflow that a non-technical regional operations team can run every quarter.

The system should support day-to-day order handling and then analyse the resulting records to surface category performance, high-value customers, fast-moving products and products at risk of stock-out.

Challenge Objectives

- Represent Customers, Products and Orders using suitable R data structures and an S3 class hierarchy.
- Implement registration, catalogue management, stock adjustment, order placement, cancellation and payment logging.
- Apply category-specific discount rules for Electronics, Fashion, Home & Kitchen, and Books & Stationery through reusable dispatch functions.
- Use data frames, filtering, sorting, merging, reshaping and CSV file I/O to maintain analytical datasets.

- Generate quarterly revenue, category, product and customer summaries using R visualisation tools.
- Provide input validation and error handling so a single bad entry cannot corrupt the dataset.
- Produce a menu-driven workflow usable by a regional operations coordinator with no R background.

Why This Challenge Requires Multiple R Techniques

The problem cannot be solved with a single calculation. Data structures hold entities, S3 classes give those entities identity and behaviour, functions modularise the business rules, data frames organise transaction-level records, file handling gives the prototype persistence across sessions, and visualisation converts rows of numbers into a decision a category manager can act on.

The workflow mirrors a regional seller-operations desk: orders drive quarterly revenue, product records drive replenishment alerts, and customer records drive an understanding of which buyers are worth retaining.

Industry and Societal Relevance

- Regional E-Commerce - enabling smaller sellers outside metro hubs to run structured digital operations.
- Multi-Category Retail Analytics - comparing Electronics, Fashion, Home & Kitchen and Books & Stationery performance side by side.
- Seller & Inventory Intelligence - flagging stock-outs before they interrupt fulfilment.
- Customer Relationship Management - separating occasional buyers from consistently high-value customers.
- Responsible Consumption - using quarterly demand signals to avoid excess inventory build-up.

C. Problem Statement

Design, implement and evaluate a menu-driven R application for a regional multi-category marketplace containing customers, products and orders.

The application shall support customer registration, catalogue management, stock updates, order placement, cancellation, category-specific discounting, payment logging and quarterly report generation.

Customers, Products and Orders shall be represented using R data structures and an S3 object-oriented design. Each product category shall carry its own discount behaviour through a common generic function rather than duplicated if/else blocks.

The application shall analyse completed transactions to calculate quarterly revenue, identify the top revenue-generating products, rank customers by spending, detect products below a defined stock threshold, and generate graphical reports.

The application shall use input validation, error handling, reusable functions, data-frame operations and CSV file I/O so the prototype stays modular and dependable at a larger seller scale.

Expected Output

- A working menu-driven R prototype.
- Structured customer, product and order data frames.
- Reusable functions and an S3 class hierarchy for Customer, Product and Order.
- Quarterly transaction summaries and category/customer reports.
- Revenue, category and product visualisations.
- Low-stock alerts and a top-revenue-product ranking.
- CSV import/export evidence and a reproducible analysis workflow.

D. Requirements and Constraints

Functional Requirements

- Register and search customers by ID or city.
- Add, update and inspect products across four categories.
- Maintain stock quantities after sales and cancellations.
- Place valid orders and record the payment mode used.
- Cancel eligible orders and reverse inventory where required.
- Apply category-specific discount rules at the point of sale.
- Calculate order totals, quarterly completed revenue and customer spending.
- Identify top-revenue products and products below the stock threshold.
- Generate graphical and tabular quarterly reports.
- Export analytical data frames to CSV files.

Performance Requirements

- Menu operations should return results without unnecessary repeated calculations.
- Reports should always be generated from the current dataset rather than hard-coded figures.
- Filtering and aggregation should remain practical if the seller base grows to thousands of records.
- Quarterly calculations should use a single, consistent definition of completed versus cancelled orders.

Technical Constraints

- The core implementation must be written in R.
- Base R data structures and data frames should be used where appropriate.
- Reusable functions must replace repeated business-rule logic.

- The solution should demonstrate an R object-oriented approach such as S3 classes.
- CSV file I/O must be included for persistence and reporting.

Time and Resource Constraints

- The prototype should run on a standard student laptop.
- The dataset should stay manageable for demonstration while the logic remains scalable.
- The implementation should prioritise correctness and clear output over interface complexity.

Reliability and Error Handling

- Reject invalid customer IDs, product IDs and non-positive quantities.
- Prevent orders when available stock is insufficient.
- Prevent duplicate or inconsistent customer/product records.
- Handle cancelled orders without letting them contribute to completed revenue.
- Validate numeric inputs and payment-mode selections.

Security, Privacy and Ethical Considerations

- Customer records should contain only the information required for the assignment (ID, name, city).
- Real payment credentials must never be stored in the R dataset.
- Customer analysis should support service improvement, not unfair treatment of any city or segment.
- Exports should avoid unnecessary personal information.
- The application should clearly distinguish analytical estimates from business decisions.

Sustainability and Responsible Consumption

- Quarterly low-stock and sales analysis can reduce emergency restocking and excess inventory for regional sellers.
- Demand visibility supports more efficient purchasing across all four categories.
- Digital quarterly reporting reduces dependence on manual, error-prone recalculation.

E. Student Work

1. Problem Understanding and Formulation

1.1 Problem Identification

The core problem is to connect the day-to-day operations of a regional multi-category marketplace with a repeatable quarterly statistical review in one R workflow.

The application must keep customer, product and order records consistent with one another while allowing the same records to be rolled up into category- and customer-level business intelligence every quarter.

1.2 Available Information and Data Model

- Customer data contains customer ID, customer name and city.
- Product data contains product ID, product name, category, price and stock.
- Order data contains order ID, customer ID, product ID, order date, quantity, unit price, payment mode and status.
- Revenue is calculated only from completed orders, net of the category-specific discount; cancelled orders are excluded from completed revenue.

1.3 Independent Dataset Used

- The independent demonstration contains 20 customers, 16 products across 4 categories, and 45 order records.
- There are 40 completed orders and 5 cancelled orders.
- Completed-order net revenue for the demonstration is ₹165,597.34.
- The mean completed order value is ₹4,139.93.
- The values are synthetic and generated specifically for academic demonstration; they are not real seller or customer transactions.

1.4 Assumptions

- Each order refers to exactly one customer and one product line.
- Unit price is stored with the order so historical order values stay stable if the catalogue price later changes.
- A completed order reduces inventory; a valid cancellation reverses the corresponding stock movement.
- The low-stock threshold is 8 units for this prototype, reflecting a smaller regional seller's stock-holding capacity.
- Payment mode is stored as a category (UPI, Card, Cash, Wallet) rather than as sensitive payment credentials.

1.5 Problem Decomposition

1. Customer management
2. Product and inventory management
3. Order processing

4. Category-specific discount rules
5. Data-frame preparation
6. Quarterly sales and customer analysis
7. Visualisation
8. File export and reporting
9. Validation

2. Application of Course Knowledge

2.1 R Data Structures

- Vectors hold simple fields such as product IDs and unit prices.
- Lists represent a flexible object bundling several related attributes before it is promoted to an S3 object.
- Data frames hold the rectangular customer, product and order datasets.
- Factors represent categorical fields such as product category, order status and payment mode.

```
# Example data-frame construction
products <- data.frame(
  product_id = c("E02", "F04", "B01"),
  product_name = c("ClearSound Buds Pro", "DenimEdge Jeans", "InkWell Notebook Set"),
  category = c("Electronics", "Fashion", "Books & Stationery"),
  price = c(2499, 1599, 399),
  stock = c(22, 25, 40)
)
```

2.2 Control Structures and Validation

- if/else conditions drive stock checks, payment validation and category discount selection.
- while loops keep the menu running until the operator chooses Exit.
- for loops support repeated per-quarter report processing when a vectorised call is less readable.
- Validation is isolated into small functions so every menu option reuses the same checks.

```
validate_quantity <- function(qty) {
  if (length(qty) != 1 || is.na(qty) || qty <= 0) {
    return(FALSE)
  }
  TRUE
}
```

2.3 Functions and Modular Programming

- `register_customer()` - creates a validated customer record.
- `add_product()` - adds a catalogue item to one of the four categories.
- `place_order()` - checks stock, applies the discount and creates an order.
- `cancel_order()` - changes status and reverses stock when the cancellation is valid.
- `apply_discount()` - dispatches the category-specific discount rule.
- `quarterly_summary()` - calculates revenue and order metrics per quarter.
- `low_stock_report()` - identifies products below the 8-unit threshold.

2.4 Object-Oriented Design in R

An S3 design represents Product, Customer and Order as classed lists, with generic functions dispatching on class so that printing, summarising and discounting all use a common interface rather than one long conditional block.

```
new_product <- function(id, name, category, price, stock) {  
  x <- list(product_id = id, product_name = name,  
            category = category, price = price, stock = stock)  
  class(x) <- c(category, "Product")  
  x  
}  
  
print.Product <- function(x, ...) {  
  cat(x$product_id, "-", x$product_name, "|", x$category,  
      "| Stock:", x$stock, "\n")  
}
```

2.5 Category-Specific Business Rules

- Electronics: 5% discount on gross amounts of ₹10,000 or more, otherwise 2% - reflecting thin margins on high-value items.
- Fashion: 10% discount above ₹3,000, otherwise 5% - encouraging bundled apparel purchases.
- Home & Kitchen: a flat 6% discount regardless of order size.
- Books & Stationery: no slab discount; the category instead competes on an everyday-low-price model.

The purpose of the rules is to demonstrate class-based dispatch of category behaviour, not to define a universal regional retail policy.

```
discount_rate <- function(category, amount) {  
  if (category == "Electronics") return(ifelse(amount >= 10000, 0.05, 0.02))  
  if (category == "Fashion") return(ifelse(amount >= 3000, 0.10, 0.05))  
  if (category == "Home & Kitchen") return(0.06)  
  0 # Books & Stationery  
}
```


2.6 Data-Frame Operations

- Filtering selects completed orders or products below the stock threshold.
- Sorting ranks customers by spending and products by revenue contribution.
- Merging joins order records with product and customer master data.
- Grouping and aggregation calculate quarterly revenue, category totals and customer totals.
- Reshaping converts quarter/category summaries into report-friendly wide tables where needed.

```
# Merge order, product and customer information
order_detail <- merge(orders, products, by = "Product_ID")
order_detail <- merge(order_detail, customers, by = "Customer_ID")

completed <- subset(order_detail, Status == "Completed")
completed$Discount_Rate <- mapply(discount_rate, completed$Category,
                                  completed$Quantity * completed$Unit_Price)
completed$Net_Revenue <- completed$Quantity * completed$Unit_Price *
  (1 - completed$Discount_Rate)
```

2.7 Correlation and Statistical Analysis

Correlation is used here to examine whether larger orders systematically attract a larger discount rate, since two of the four category rules are quantity/amount-slab based.

```
cor(completed$Quantity, completed$Discount_Rate, use = "complete.obs")
# Independent dataset result: approximately 0.19 (weak positive association)
```

A weak positive correlation is expected: only Electronics and Fashion carry slab-based discounts, while Home & Kitchen is flat and Books & Stationery has none, so quantity does not uniformly drive the discount rate across the whole dataset.

2.8 Regression Model

A multiple linear model estimates completed net revenue from quantity, unit price and the discount rate applied, giving a simple analytical baseline rather than a production forecasting model.

```
fit <- lm(Net_Revenue ~ Quantity + Unit_Price + Discount_Rate, data = completed)
summary(fit)

# Prediction for a hypothetical order
new_order <- data.frame(Quantity = 2, Unit_Price = 2799, Discount_Rate = 0.06)
predict(fit, newdata = new_order)
```

2.9 Decision Tree Concept

A decision tree can classify each product as Fast-Moving or Slow-Moving using variables such as price, stock and category, giving the regional operations team an interpretable rule set rather than a single ranked list. The tree structure supports quick rule discovery but should be validated on a larger seller-wide dataset before it is used operationally.

3. Solution / Design / Methodology

3.1 Overall Architecture

10. Input / menu layer
11. Customer and product storage
12. Order transaction layer
13. Category-specific validation and discount rules
14. Data-frame transformation
15. Quarterly statistical analysis
16. Visualisation
17. CSV report export

3.2 Order Processing Logic

18. Read customer ID, product ID and quantity.
19. Validate that the customer and product exist.
20. Check that the quantity is positive.
21. Check available stock against the requested quantity.
22. Determine the category and dispatch the applicable discount rule.
23. Calculate the gross amount, discount and net revenue.
24. Record the payment mode and order status.
25. Reduce stock by the ordered quantity.
26. Return a transaction confirmation to the operator.

3.3 Cancellation Logic

27. Find the order by order ID.
28. Check that it is currently completed and eligible for cancellation.
29. Change status to Cancelled.
30. Restore the sold quantity to inventory.
31. Exclude the cancelled order from all completed-revenue reports.

3.4 Report Design

- Dashboard metrics: quarterly completed revenue, completed orders, registered customers and low-stock product count.
- Category report: revenue contribution by Electronics, Fashion, Home & Kitchen and Books & Stationery.
- Product report: revenue contribution and remaining stock per product.

- Customer report: total spending and completed-order frequency.
- Order report: transaction-level detail with status and payment mode.

3.5 Menu Structure

32. Customer Management
33. Product Management
34. Place Order
35. Cancel Order
36. Inventory Status
37. Quarterly Sales Analytics
38. Customer Analytics
39. Export Reports
40. Exit

4. Use of Modern Tools

4.1 R Tools and Packages

- R - core programming, data structures, functions and statistical modelling.
- RStudio - script development, console execution and project organisation.
- ggplot2 - the quarterly, category and product visualisations used in this report.
- dplyr - grouping, summarising and joining the order, product and customer tables.
- tidyr - reshaping quarter/category summaries into report-friendly structures.
- readr - CSV import and export for the customer, product and order tables.
- rpart / rpart.plot - the Fast-Moving versus Slow-Moving product classification described in Section 2.9.

4.2 Evidence of Tool Usage

The code blocks in this assignment are written in R and correspond to the independent dataset and workflow described above; the figures in Section 5 were produced from the same aggregated calculations.

A supplied reference application was used only as a functional-workflow reference point (dashboard metrics, order handling, catalogue and stock monitoring, customer intelligence, analytics and report export). The dataset, category set, discount rules, wording, charts and R logic in this submission are independently constructed for this team and are not copied from that reference or from any other team's submission.

4.3 File I/O

```
write.csv(customers, "customers.csv", row.names = FALSE)
write.csv(products, "products.csv", row.names = FALSE)
write.csv(orders, "orders.csv", row.names = FALSE)

customers_in <- read.csv("customers.csv")
```

```
products_in <- read.csv("products.csv")
orders_in <- read.csv("orders.csv")
```

4.4 Visualisation Code

```
library(ggplot2)
library(dplyr)

quarterly <- completed %>%
  mutate(Quarter = paste0("Q", quarters(as.Date(Order_Date)))) %>%
  group_by(Quarter) %>%
  summarise(Revenue = sum(Net_Revenue))

ggplot(quarterly, aes(Quarter, Revenue)) +
  geom_col(fill = "#8B1E1E") +
  labs(title = "Quarterly Completed Revenue",
       x = "Quarter", y = "Revenue (\u20b9)") +
  theme_minimal()
```

5. Results and Validation

5.1 Overall Transaction Results

Metric	Independent Result
Registered customers	20
Products in catalogue	16
Total order records	45
Completed orders	40
Cancelled orders	5
Completed net revenue	₹165,597.34
Average completed order value	₹4,139.93
Low-stock products (< 8 units)	5

5.2 Quarterly Revenue

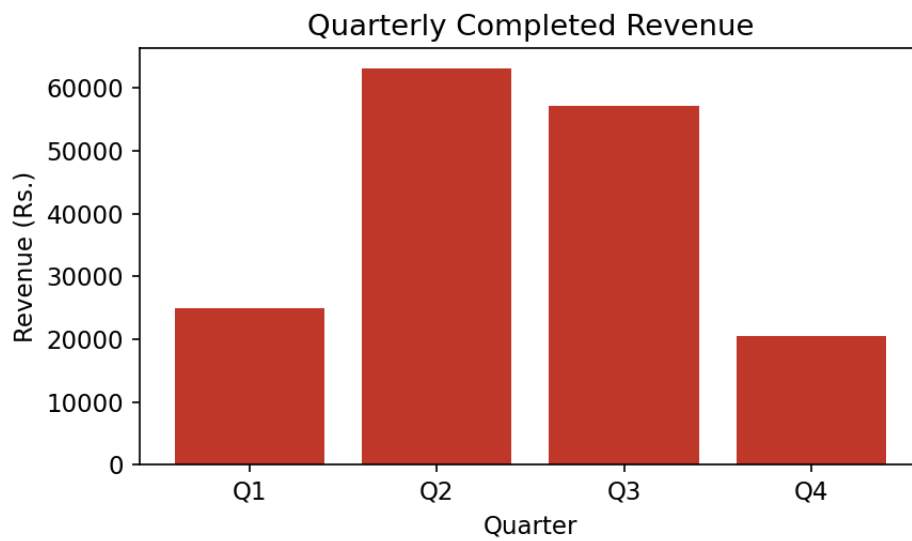


Figure 5.1 - Independent quarterly completed-revenue analysis.

5.3 Category Performance

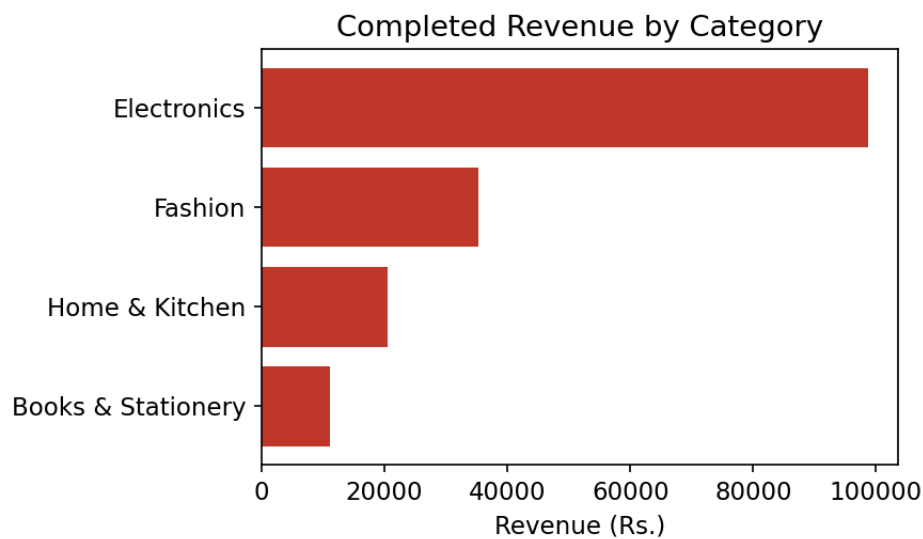


Figure 5.2 - Independent completed revenue by product category.

5.4 Top Revenue-Generating Products

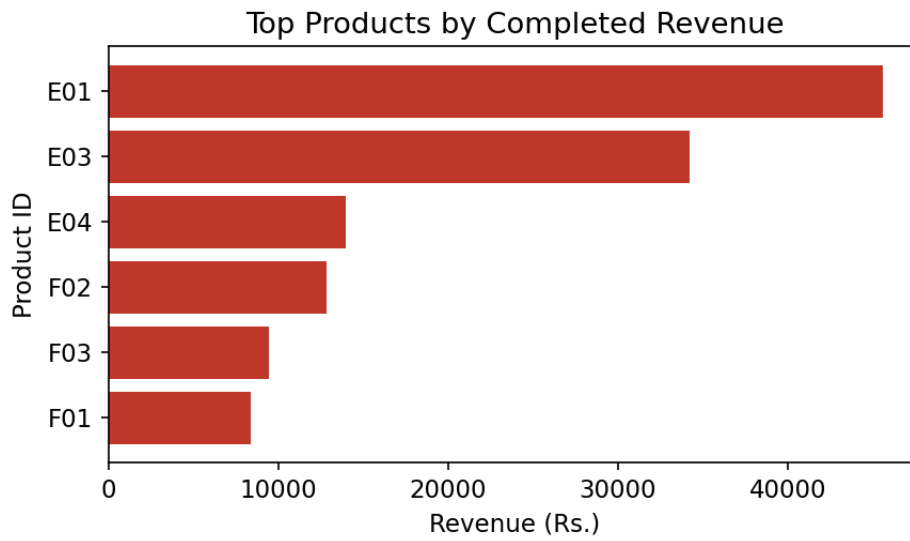


Figure 5.3 - Independent product ranking by completed revenue.

5.5 Low-Stock Analysis

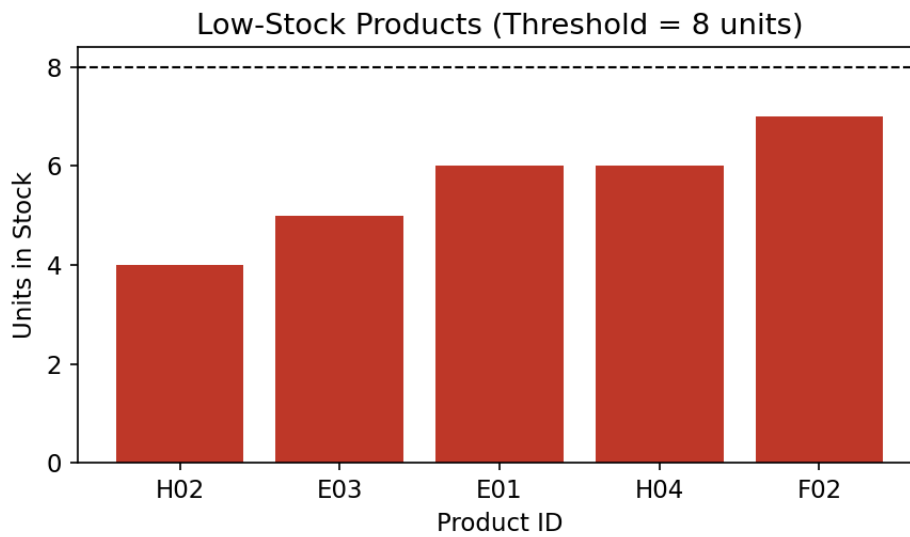


Figure 5.4 - Independent low-stock monitoring output (threshold = 8 units).

5.6 Quantity and Discount Relationship

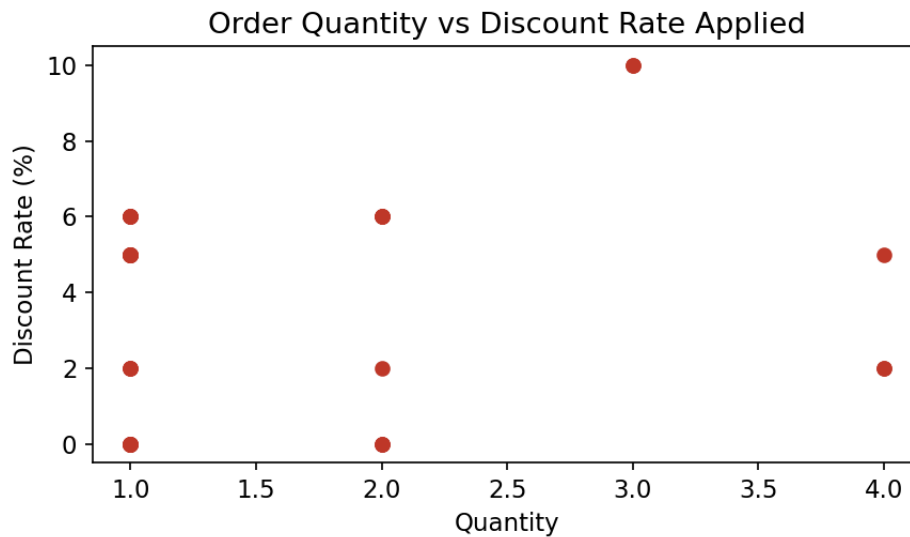


Figure 5.5 - Independent relationship between order quantity and the discount rate applied.

5.7 Validation Checks

- Cancelled orders are excluded from completed revenue.
- Every completed order references a customer and product present in the respective master data.
- Net revenue is calculated as quantity $\times$ unit price, less the category-specific discount.
- Low-stock reporting consistently uses the defined threshold of 8 units.
- Customer totals are calculated from completed orders only.
- Product rankings use completed-order revenue rather than cancelled-order figures.
- CSV exports preserve the analytical fields required for later reloading.

5.8 Sample Customer Ranking

Customer_ID	Total_Spending (₹)	Completed_Orders
C17	50,691.13	2
C05	35,715.25	2
C19	12,244.30	3
C14	11,526.42	3
C15	9,219.16	6
C13	7,522.14	4
C20	6,959.14	2

Customer_ID	Total_Spending (₹)	Completed_Orders
C07	5,147.13	3

5.9 Low-Stock Table

Product_ID	Product_Name	Category	Stock
H02	GlowLite Study Lamp	Home & Kitchen	4
E03	PixelView Monitor 22	Electronics	5
E01	SwiftBook Air 13	Electronics	6
H04	ChopEase Blender	Home & Kitchen	6
F02	StrideFit Sneakers	Fashion	7

6. Analysis and Engineering Decision

6.1 Interpretation of Results

The independent dataset produces ₹165,597.34 in completed net revenue from 40 completed transactions across four categories.

The quarterly chart shows revenue concentrated in Q2 and Q3, which would prompt a regional operations team to check whether that pattern reflects a genuine seasonal cycle or a scheduling gap in Q1 and Q4 seller activity.

The category chart shows Electronics contributing the largest share of revenue despite carrying the smallest discount rates, while Fashion contributes strongly through its higher slab-based discount tier - together these suggest category-specific promotion budgets rather than a single blanket discount policy.

The top-product-by-revenue view highlights which SKUs justify prioritised restocking, while the low-stock chart flags five products that need replenishment attention before the next quarter.

Customer ranking turns transaction-level records into customer-value information: a small number of customers (C17, C05) account for a disproportionate share of spending, which supports a targeted retention approach rather than a uniform one.

6.2 Data-Frame Approach versus Separate Objects

Data frames are used for analytical reporting because filtering, joining and aggregation across customers, products and orders are straightforward and map directly to how a quarterly report is structured.

S3 objects are used for domain behaviour - particularly for category-specific discount dispatch - because they let the same generic function call resolve to different business logic per category without repeating conditional blocks throughout the codebase.

Combining both representations is preferable to forcing every operation into a single one: data frames for reporting, S3 objects for behaviour.

6.3 Base R versus Tidyverse

- Base R is sufficient for the core data structures, functions, control flow and statistical models used here.
- dplyr makes the repeated per-quarter and per-category grouping and joining easier to read and maintain.
- tidyr supports reshaping quarterly summaries into report-friendly wide tables where needed.
- ggplot2 gives the five figures in Section 5 a consistent, presentation-ready style.
- The final implementation mixes base R and tidyverse tools as long as the workflow stays reproducible.

6.4 Advantages

- Modular functions avoid duplicating the discount and validation logic across menu options.
- Data frames support structured, repeatable quarterly analytics.
- S3 classes demonstrate genuine object-oriented dispatch rather than a single long conditional.
- CSV I/O makes the prototype portable between sessions and machines.
- The five visual reports make quarterly and category patterns easy to interpret at a glance.
- Validation reduces the risk of the dataset entering an inconsistent state.

6.5 Limitations

- The dataset is synthetic and relatively small (45 orders across 20 customers and 16 products).
- The prototype does not connect to a live payment gateway or a production database.
- The four discount rules are illustrative of category-based dispatch, not a universal regional retail policy.
- The regression and decision-tree components are analytical demonstrations and would need a larger, seller-wide dataset for reliable forecasting.
- A production application would need authentication, database transactions, audit logs and stronger security controls.

6.6 Engineering Decision

The selected design is a menu-driven R application backed by structured data frames, an S3 class hierarchy for Customer, Product and Order, reusable discount-dispatch functions, and CSV persistence.

For analytics, a tidy quarterly workflow with grouped summaries and ggplot2 visualisations is preferred because it keeps every transformation step traceable back to the raw order records.

This architecture satisfies the assignment's functional and analytical requirements while leaving a clear path toward a real database backend and a graphical interface for regional operations staff.

7. Broader Considerations

7.1 Sustainability

- Quarterly low-stock and category analytics can reduce unnecessary overstocking for smaller regional sellers with limited storage capacity.
- Digital reporting removes the need for repetitive manual, paper-based stock and sales reconciliation.

7.2 Society and Customers

- Category and customer analytics can improve product availability and service quality across Tier-2/3 markets.
- Customer-level insight should not be used to unfairly exclude or disadvantage buyers from any particular city or segment.

7.3 Safety and Reliability

- Incorrect stock updates could cause overselling, so order and cancellation logic must keep inventory consistent at every step.
- Input validation and status checks are essential before a regional operations team can trust the system's numbers.

7.4 Ethics and Privacy

- Only the minimum customer information (ID, name, city) is retained for analytics.
- Payment credentials are never stored in the assignment dataset.
- Customer segmentation should be periodically reviewed for unfair or discriminatory outcomes.

7.5 Economics

- Quarterly revenue, top-product and low-stock reports support purchasing, promotion and staffing decisions for regional sellers.
- Automating repetitive calculations reduces the operational effort needed to run each quarterly review.

7.6 Accessibility and Professional Responsibility

- Reports use readable labels, clear tables and simple charts so a non-technical operations coordinator can act on them.
- Developers are responsible for documenting assumptions and limitations rather than presenting synthetic demonstration data as real business evidence.

8. Conclusion

A complete regional, multi-category e-commerce analytics problem can be addressed by combining R programming fundamentals with structured data management, an S3 object-oriented design and quarterly statistical analysis.

The independent prototype models customers, products and orders across four categories, supports order and cancellation logic, applies category-specific discount rules, and generates quarterly revenue, category, product and customer insights.

The results show how R can move from raw, per-order transaction records to a quarterly view of revenue, category performance, high-value customers and stock risk.

The design satisfies the assignment's major functional and analytical requirements while remaining a reasonable starting point for a future database backend and a graphical interface for a regional operations team.

The main limitation is that the demonstration uses a small synthetic dataset; a real regional deployment would need a larger transaction history, stronger security controls, testing and business validation before use.

9. Student Reflection

9.1 Learning Beyond Direct Classroom Theory

- Working through a full quarterly reporting cycle showed how individual R techniques (vectors, control flow, functions, S3 classes) combine into one dependable workflow rather than staying isolated exercises.
- Separating the raw order records from the derived quarterly summaries made it clear why reports should always be recalculated from source data rather than stored as fixed numbers.
- Designing category-specific discount rules as S3-dispatched functions showed a concrete benefit of object-oriented design over a long chain of if/else statements.
- Building the five visual reports showed how the same statistical calculation can be presented at different levels - quarter, category, product and customer - for different decision-makers.

9.2 Challenges Faced

- Keeping customer, product and order records consistent with one another after every order and cancellation.
- Correctly excluding cancelled orders from every revenue, category and customer calculation.
- Designing discount and validation functions that could be reused across menu options without duplication.
- Choosing chart types that communicate the quarterly and category patterns without overcrowding the report.

9.3 Skills Developed

- R data structures and data-frame manipulation.
- Functions, control structures and input validation.
- S3 object-oriented programming and generic-function dispatch.
- Data merging, grouping and aggregation across three related tables.
- CSV file handling for persistence and reporting.
- Correlation analysis and simple linear regression.
- Decision-tree concepts for interpretable product classification.
- Business-oriented data visualisation with ggplot2.

9.4 Improvements With Additional Time and Resources

- Connect the R application to SQLite or PostgreSQL for persistent, multi-seller data storage.

- Add authentication and role-based access for regional administrators versus individual sellers.
- Build a Shiny dashboard with live quarterly KPIs, category filters and downloadable reports.
- Add simple time-series forecasting for category-level demand ahead of each quarter.
- Introduce automated unit tests for the order, cancellation and discount-dispatch functions.
- Use a larger, properly licensed and anonymised regional dataset for stronger statistical validation.

10. References

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F. Common Assessment Rubric

Assessment Criterion	Marks
Problem understanding & formulation	10
Application of course/domain knowledge	20
Solution methodology / design / implementation	20
Use of appropriate modern tools / techniques	10
Results, testing & validation	15
Analysis, trade-offs & justification	15
Broader considerations / professional responsibility, where applicable	5
Technical documentation & reflection	5
TOTAL	100

G. CO-PO-Assessment Mapping

Note: Faculty shall map only the POs and Bloom's levels genuinely assessed by this particular assignment. Every assignment is not required to map to every PO.

Assessment Component	CO	PO(s)	Bloom's Level	Marks
Problem formulation			L3/L4	10
Application of knowledge			L3/L4	20
Solution / Design / Implementation			L4/L5/L6	20
Modern tool usage			L3/L4	10
Validation			L4/L5	15
Analysis & justification			L4/L5	15
Broader considerations			L4/L5	5
Documentation & reflection			L3/L4	5

H. Faculty Evaluation & Continuous Improvement

CO Attainment

Performance Level	Criterion
Level 3	$\geq 70\%$
Level 2	60-69%
Level 1	50-59%
Level 0	$< 50\%$

Target CO Attainment:

Actual CO Attainment:

Faculty Analysis

Areas in which students performed well:

Areas requiring improvement:

Corrective / Improvement Action:

Action to be implemented in the next offering:

Documents to be Maintained

The department/course faculty shall maintain:

41. Assignment question/problem statement
42. CO-PO and Bloom's mapping
43. Assessment rubric
44. Student submissions
45. Tool/simulation/experimental evidence, where applicable
46. Evaluated scripts/reports
47. Marks and rubric-wise assessment
48. CO attainment analysis
49. Sample student work representing different performance levels
50. Faculty analysis and continuous-improvement action

```
275     ylab = "revenue"
276   )
277
278 } else if (choice == "7") {
279
280   cat("\nCorrelation:",
281       round(correlation, 3), "\n")
282
283   print(summary(model))
284
285 } else if (choice == "0") {
286
287   cat("\nThank you for using the system!\n")
288   break
289
290 }
```

295:1 # (Untitled) R Script

Console Terminal Background Jobs

```
R 4.5.1 ~/  
+ print(summary(model))  
+  
+ } else if (choice == "0") {  
+  
+   cat("\nThank you for using the system!\n")  
+   break  
+  
+ } else {  
+  
+   cat("\nInvalid choice. Try again.\n")  
+ }  
+ }
```

===== E-COMMERCE SYSTEM =====

1. View Customers
2. View Products
3. Place Order
4. Search Customer by City
5. Inventory Status
6. Sales Analysis
7. Statistical Analysis
0. Exit

Enter your choice: |